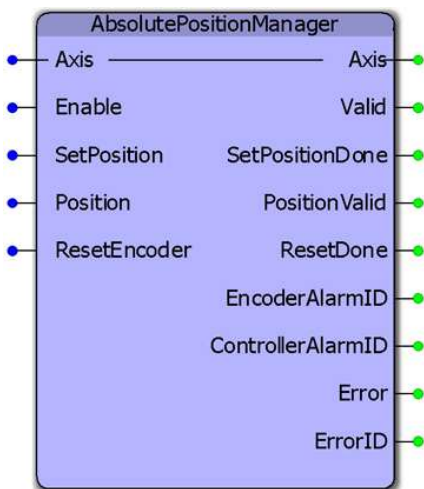




AbsolutePositionManager



This function monitors for any controller or servo alarm related to the absolute encoder or battery backed encoder offset data stored in the controller. It can serve as the single point of monitoring, clearing, and defining the position of an absolute encoder. This function includes a retained Boolean output variable that once set, requires that the alarm be cleared through this function, and that the position of the encoder is redefined. The intention is to prevent the machine from operating until the position of the absolute encoder has been calibrated to the machine coordinates.

This function includes the following PLCopen function blocks: MC_ReadAxisError, MC_ReadAlarm, MC_ResetAbsoluteEncoder, Y_ClearAlarm and MC_SetPosition.

Library

PLCopen Toolbox

Parameters

*	Parameter	Data Type	Description
VAR_IN_OUT			
B	Axis	AXIS REF	Logical axis reference. This value can be located on the Configuration tab in the Hardware Configuration (logical axis number).
VAR_INPUT			Default
B	Enable	BOOL	The function will continue to execute every scan while Enable is held high and there are no errors.
V	SetPosition	BOOL	Value of the axis position to be set when homing is Done.

V	Position	LREAL	A positive or negative value within the coordinate system in user units.	LREAL#0.0
V	ResetEncoder	BOOL	Initiates the Y_ResetAbsoluteEncoder function to clear any absolute encoder related SERVOPACK alarm, including A.810 and A.CC0.	LREAL#0.0
VAR_OUTPUT				
B	Valid	BOOL	Indicates that the function is operating normally and the outputs of the function are valid.	
V	SetPositionDone	BOOL	Indicates that MC_SetPosition has successfully completed.	
V	PositionValid	BOOL	Indicates that the absolute encoder has no alarms, and the MC_SetPosition has been used at some point in the past to align the encoder with the mechanical system.	
V	ResetDone	BOOL	Indicates that the ResetEncoder request has completed successfully.	
V	EncoderAlarmID	UINT	ServoPack alarm related to the absolute encoder.	
V	ControllerAlarmID	UDINT	Controller alarm related to the SRAM or battery, which stores the absolute encoder offset.	
B	Error	BOOL	Set high if an error has occurred during the execution of the function block. This output is cleared when 'Execute' or 'Enable' goes low.	
B	ErrorID	UINT	If Error is true, this output provides the Error ID. This output is reset when 'Execute' or 'Enable' goes low.	

Notes

- The PROCONOS firmware library is required in the project when using this function block due to its use of some additional functionality provided there.
- This function block is not supported on the Sigma7-Siec because there is a Retain variable in the code which is not supported due to hardware limitations; there is no SRAM memory on the Sigma7-Siec.
- To clear the absolute encoder alarm from a ServoPack, use the 'ResetEncoder' input, then reset the ServoPack by either cycling power to the ServoPack or using the Y_ResetMechatrolink function block.
- Check the Hardware Configuration to ensure that the alarm format for Sigma III and higher drives is set for 3 digit alarm mode.
- See the [AbsolutePositionManager eLearning Module](#) on Yaskawa's YouTube channel.

Error Description

See the [Function Block ErrorID](#) list.